



Full Stack Development

Industrial Training Kit

Project 3

Batch: 2026 | Powered by DecodeLabs



WELCOME TO THE TEAM!

Step into the role of a Full Stack Developer at DecodeLabs. Project 3 is your persistence phase: Database Integration. This track isn't about "temporary variables"—it's about Data Longevity. Before you build complex user authentication, you must master the art of designing schemas and performing CRUD (Create, Read, Update, Delete) operations to manage information effectively. By completing this milestone, you are proving you can bridge the gap between application logic and permanent storage through pure architectural logic. Let's design a digital vault that handles data with absolute reliability.



GUIDELINES:

Qualification Criteria

EARN YOUR BADGE 🛡️

- Requirement: Complete project 3 to unlock other projects for next week.
- Standard: All projects must be verified for quality.

Mandatory: Project 3 is the definitive milestone for mastering your domain.



Project 3:

Database Integration



Goal:

Connect the backend with a database to store and retrieve data.

Key Requirements:

- Design a simple database schema
- Perform basic CRUD operations
- Ensure proper data handling

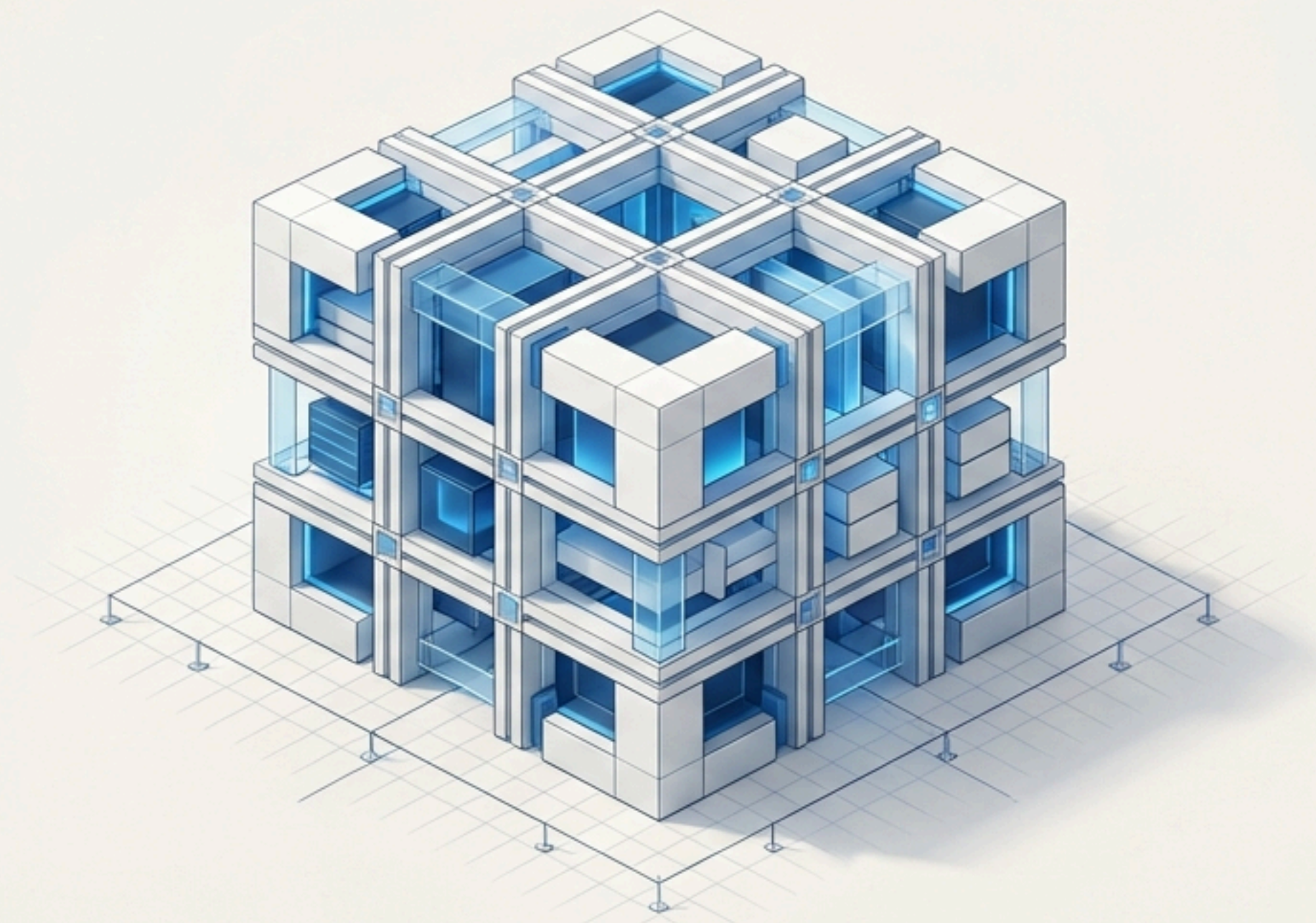
Key Skills:

Databases, CRUD operations, data storage

Project 3

Database Integration

Mastering the Architecture of State Persistence



Conquering the Gatekeeper to Backend Mastery



Pillar 1: The Blueprint

Schema & Design

Pillar 2: The Bridge

Integration & Connection

Pillar 3: The Action

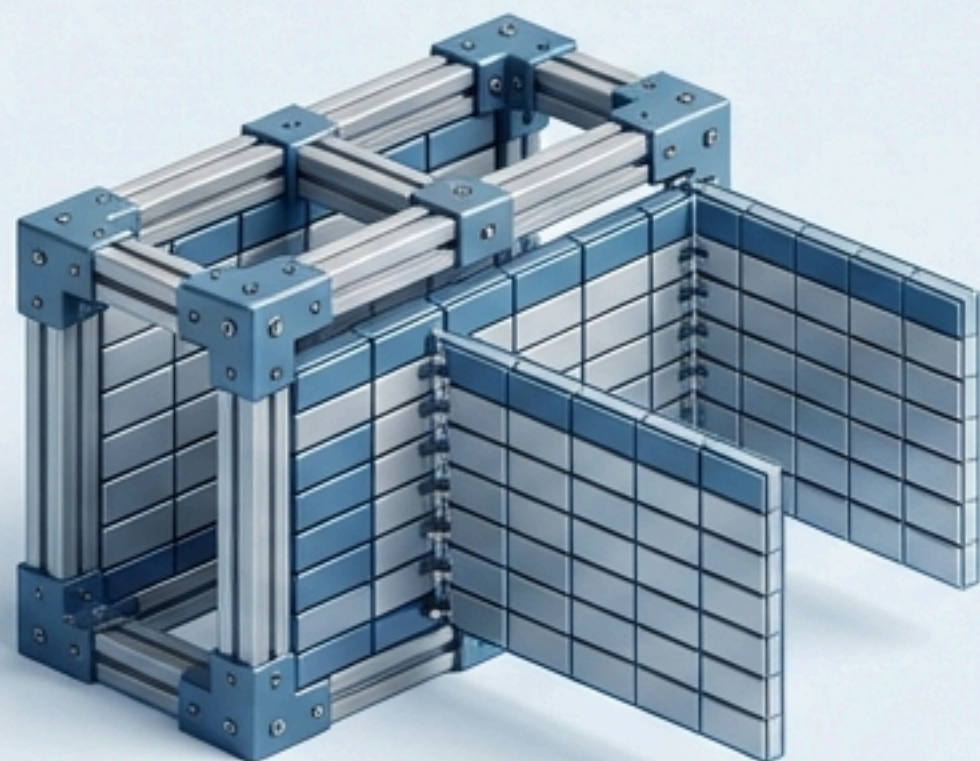
CRUD & RESTful HTTP

Pillar 4: The Shield

Integrity & Security

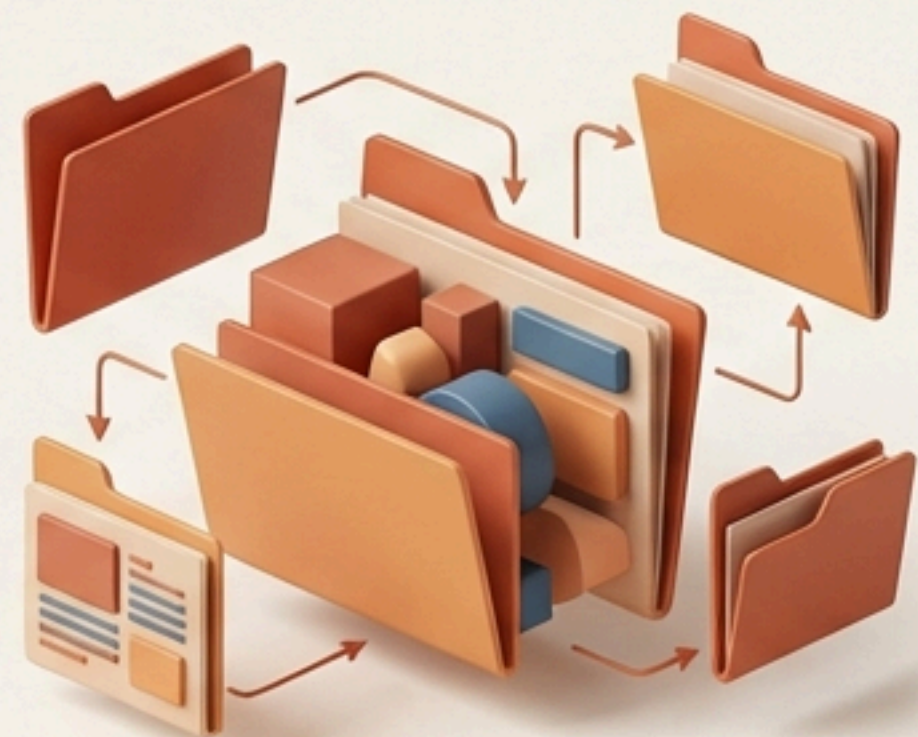
You cannot interact with data until you connect to it. You cannot connect until you design it. Once it is built, you must protect it.

Choosing the Right Blueprint for Your Data



SQL (Relational)

- Strictly structured, tabular format
- Predictable and ACID-compliant
- Ideal for complex relationships
- Examples: PostgreSQL, MySQL



NoSQL (Non-Relational)

- Flexible, schema-less document design
- Highly scalable horizontally
- Ideal for rapidly changing datasets
- Example: MongoDB

Structuring Data Through Relational Geometry



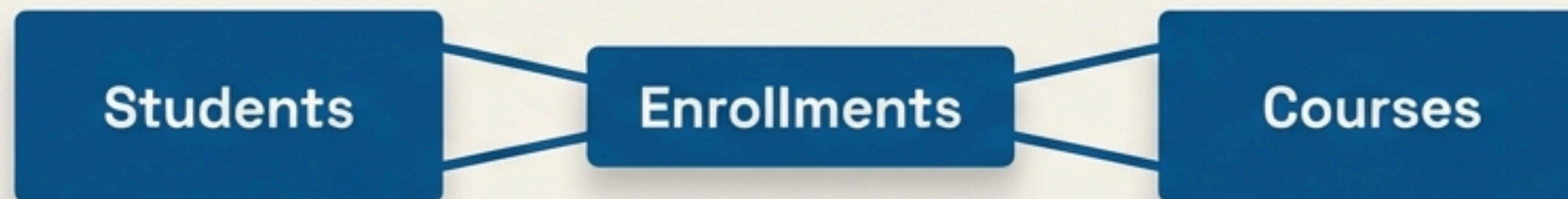
One-to-One (1:1)

A strictly exclusive connection. A User record connects to exactly one UserProfile record.



One-to-Many (1:Many)

The most common relationship. A single Customer record can connect to multiple Order records.



Many-to-Many (M:M)

Requires a junction table. Multiple Students enroll in multiple Courses, mapped through a dedicated intermediary table.

Binding the System Together with Keys

Primary Key: The unique identifier for every single row in a table. Ensures absolute distinctness. No two records are identical.

Users

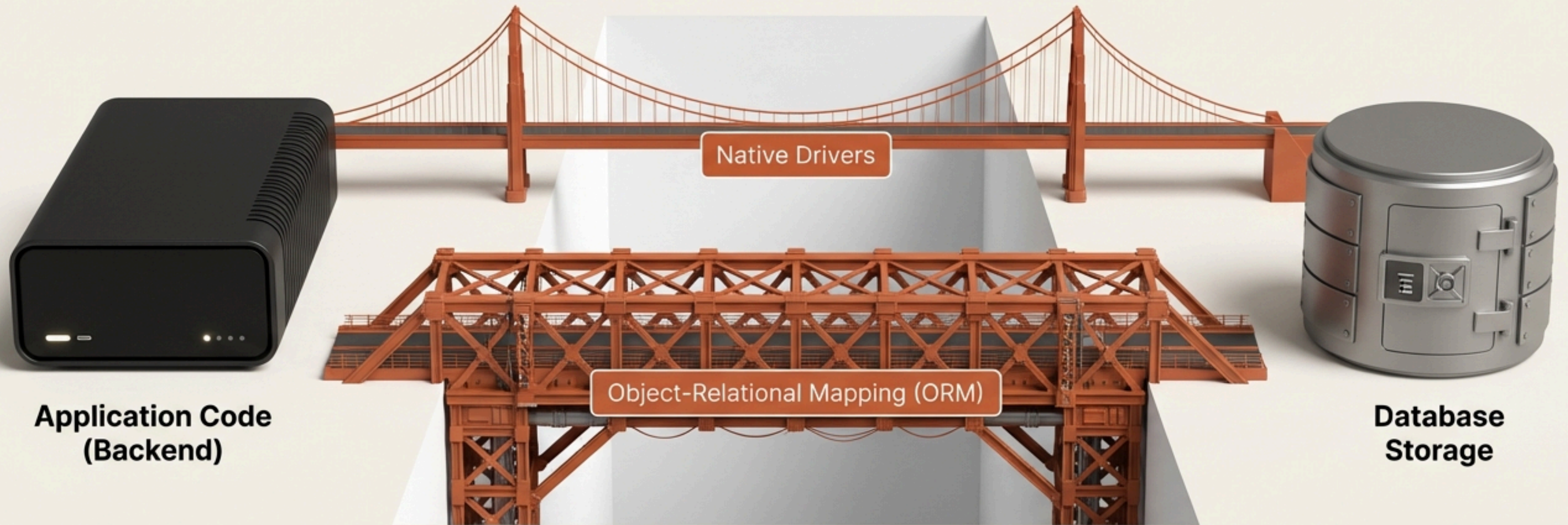
UserID 	Name	Email
101	Alice	alice@email.com

Orders

OrderID	Total	UserID 
505	\$42.00	101

Foreign Key: The structural glue. It points directly back to a Primary Key in another table, establishing a relationship and maintaining referential integrity.

Bridging the Gap Between Code and Storage



Application code requires a dedicated translation layer to communicate with a database. Your architecture depends entirely on which bridge you choose to build.

Evaluating Native Drivers Against ORMs

Native Drivers

Example: pg for Postgres

- ✓ Requires writing raw SQL queries
- ✓ Operates at maximum execution speed
- ✓ Requires repetitive boilerplate code
- ✓ Involves manual mapping of rows to variables

ORMs

Examples: Prisma, SQLAlchemy, Mongoose

- ✓ Translates database rows into object-oriented code
- ✓ Provides automatic type-safety and autocompletion
- ✓ Massively increases developer productivity
- ✓ Introduces a learning curve and slight performance overhead

The Fundamental Operations of State Persistence



Mapping Database Actions to the Web

CREATE = HTTP POST = SQL INSERT

POST is non-idempotent; identical requests create multiple new resources.

READ = HTTP GET = SQL SELECT

UPDATE = HTTP PUT = SQL UPDATE

(or PATCH for partial updates)

DELETE = HTTP DELETE = SQL DELETE

The Database is the Final Source of Truth

Databases must never trust application logic blindly. Data integrity must be enforced strictly at the schema level.

UNIQUE

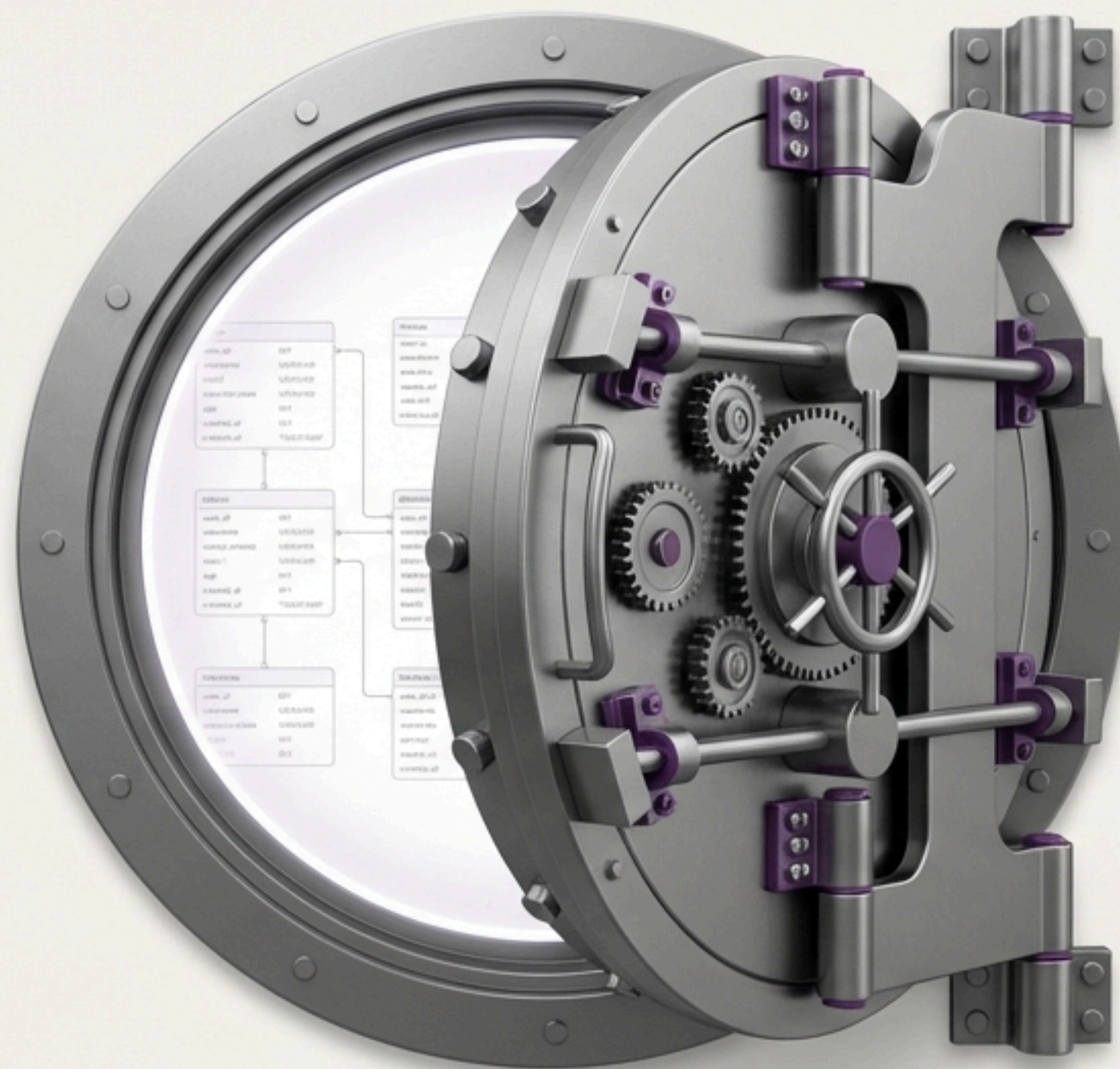
Enforces that all values in a column are entirely distinct. Prevents duplicate usernames or emails.

NOT NULL

Ensures a column can never be left empty. Critical data points must always be provided.

CHECK

Imposes a strict mathematical or logical condition on the exact values allowed in a column (e.g., $\text{age} \geq 18$).



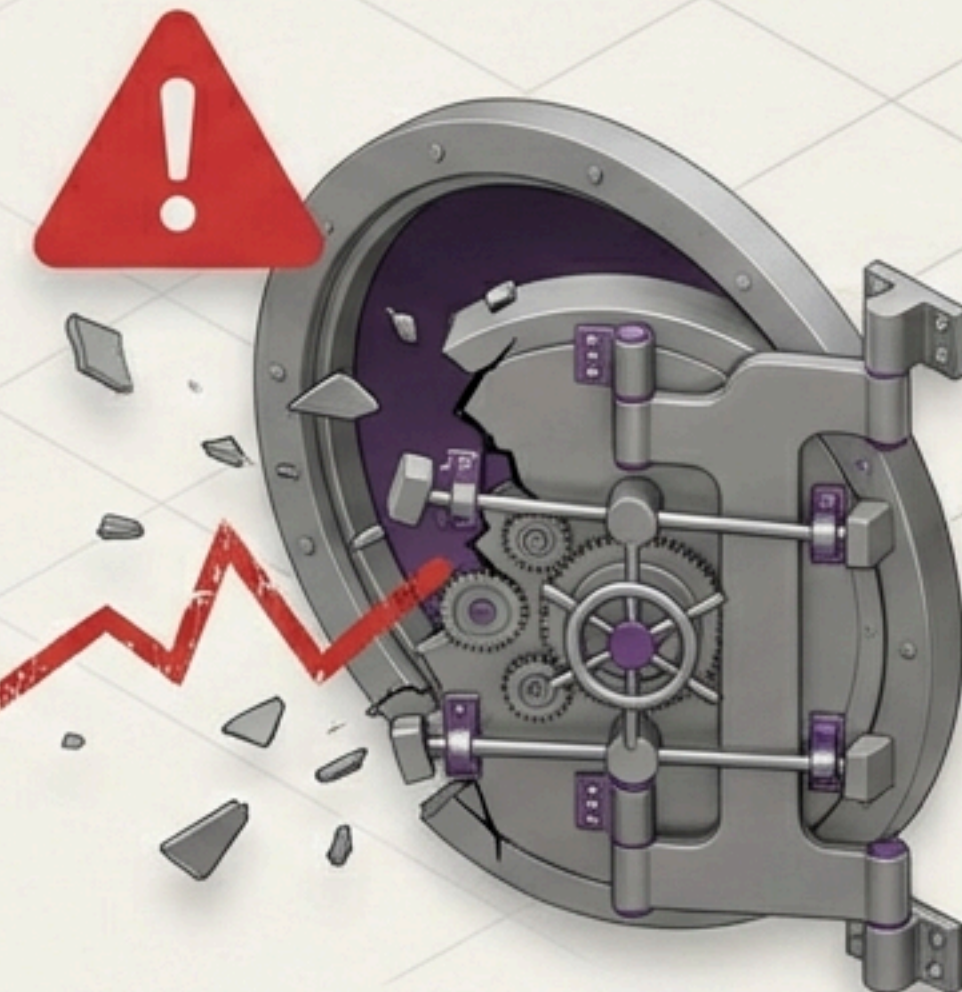
The Persistent Threat of SQL Injection

Email Address

' OR 1=1 --

Malicious
User Input

```
SELECT * FROM users  
WHERE email = '' + userInput + '';
```



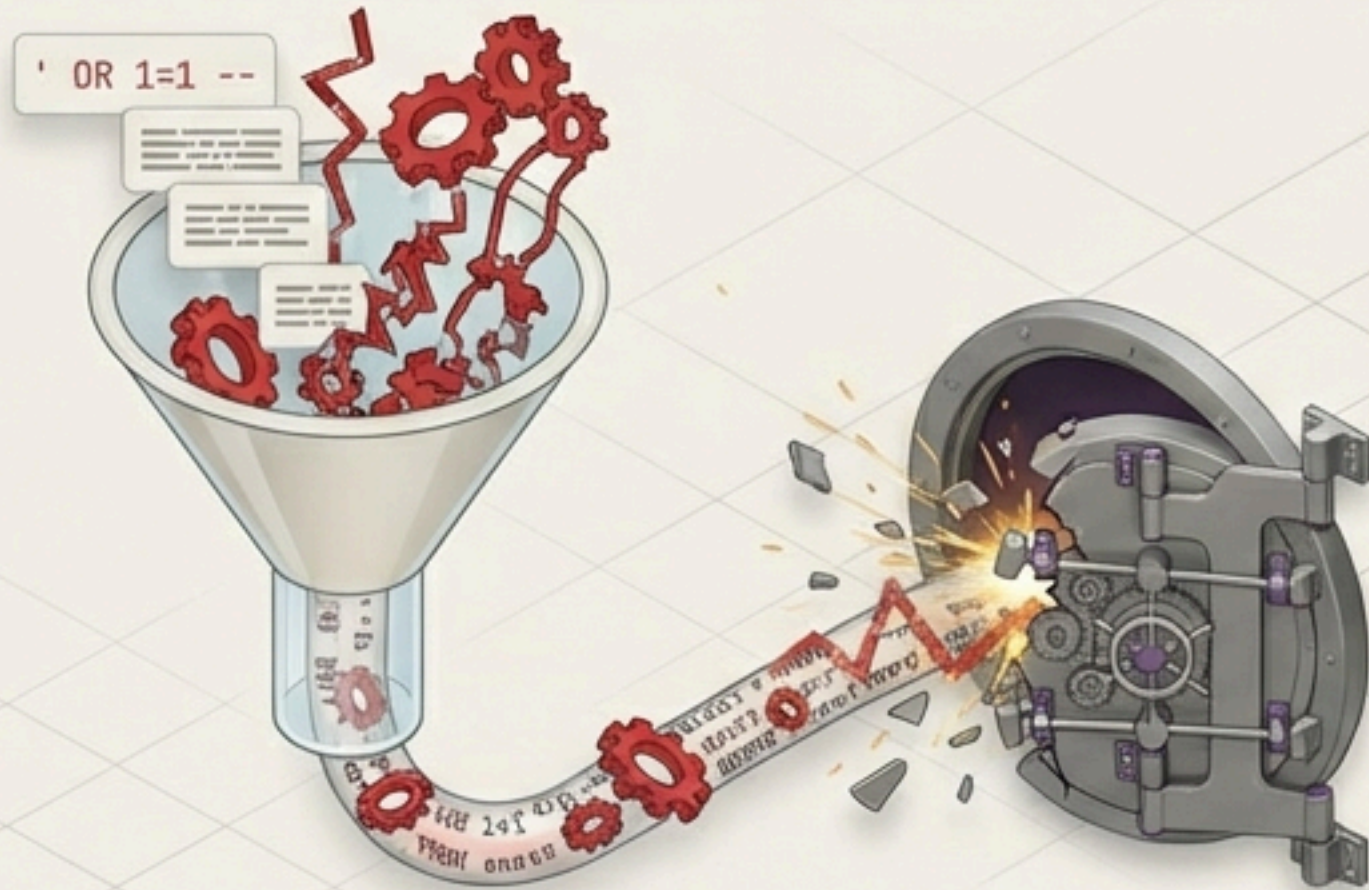
```
SELECT * FROM users  
WHERE email = '' OR 1=1 --';
```

Raw concatenation fails to distinguish between intended data and malicious executable logic, allowing attackers to bypass authentication entirely.

Neutralizing Attacks with Parameterized Queries

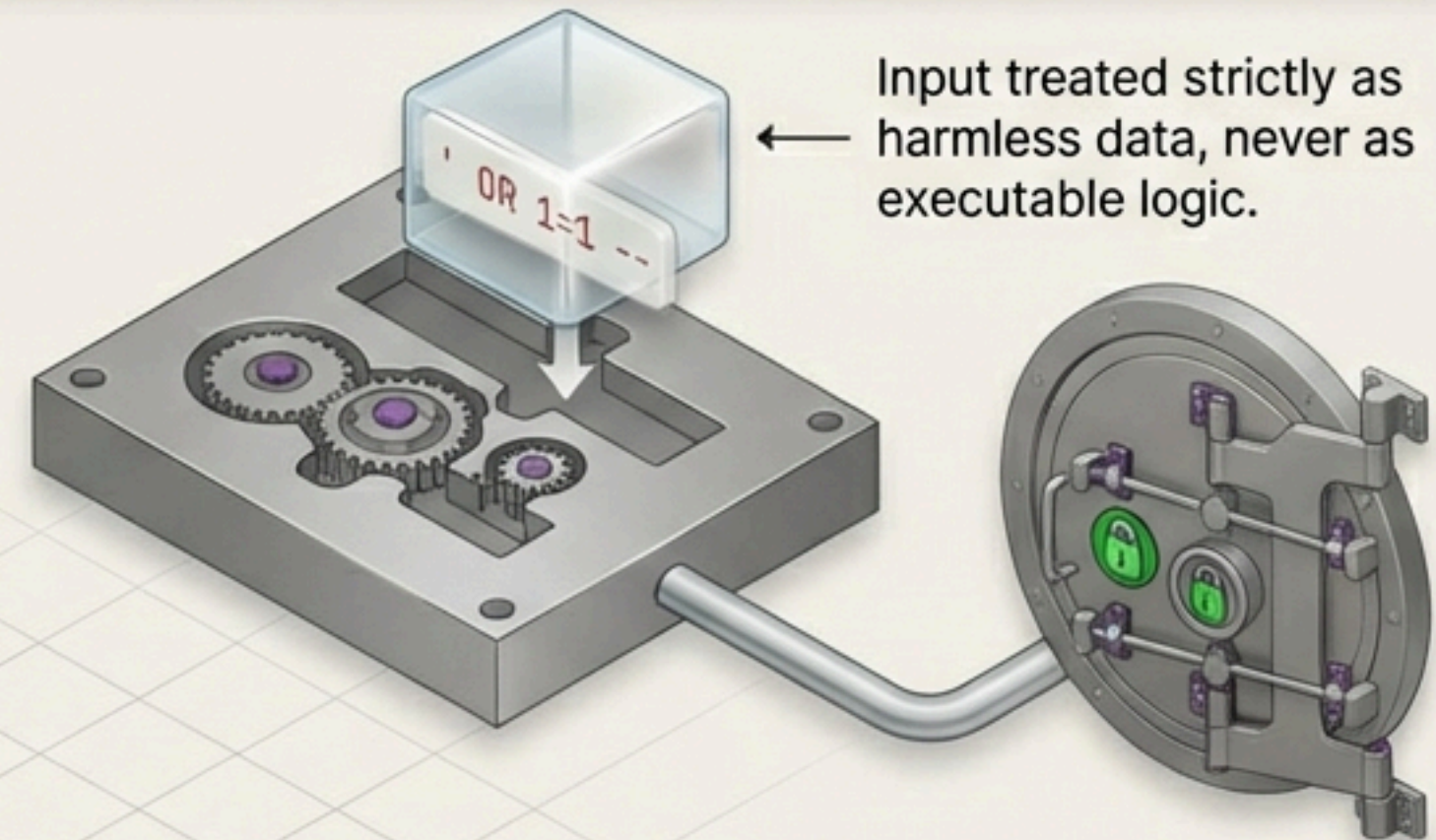
Vulnerable: String Concatenation

```
query = 'SELECT * FROM users WHERE email = ' + userInput;
```

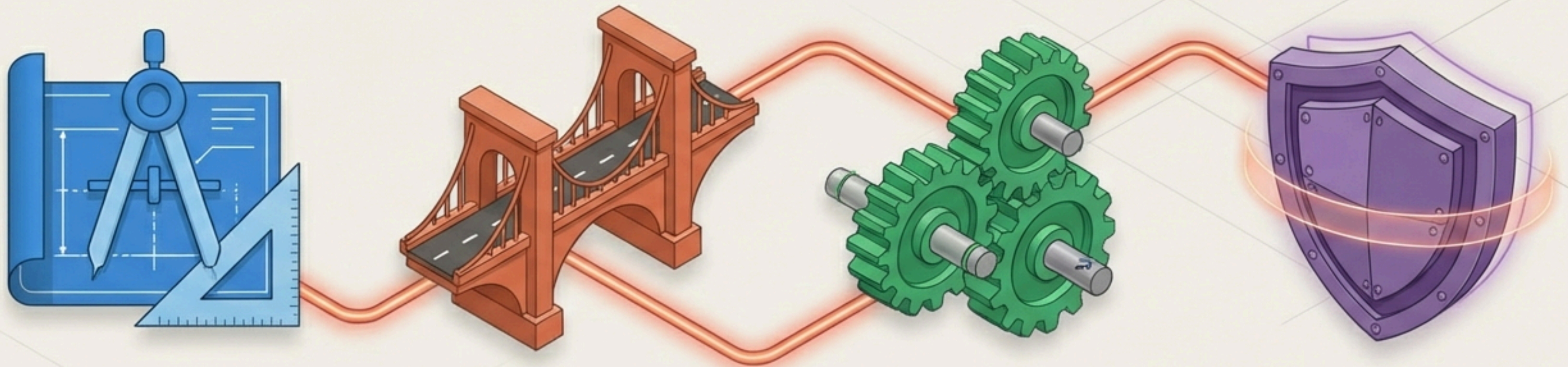


Secure: Parameterized Queries

```
query = 'SELECT * FROM users WHERE email = $1';  
db.execute(query, [userInput]);
```



Enter the Architecture of State Persistence



Design your schema with precision.

Bridge the gap with the right integration tool.

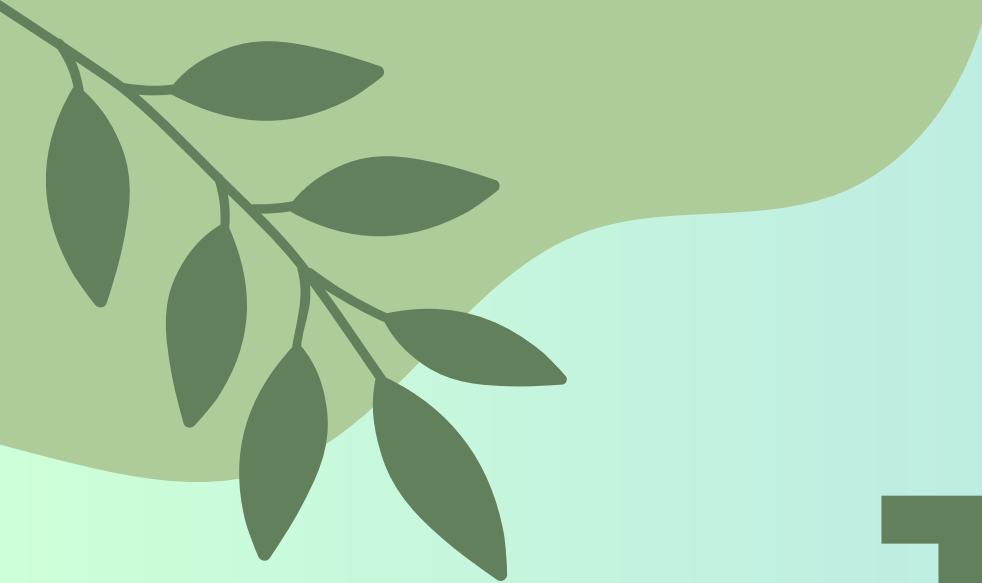
Map your CRUD operations to RESTful standards.

Shield your data with strict constraints and parameterization.

Project 3 is ready for you. Build your database integration.

CONCLUSION

The absolute best way to master Full Stack Development is through hands-on practice, not just theory. Don't just aim to complete these projects; take them one by one, experiment with unique solutions—like optimizing your queries or adding data relationships—and treat every "Database Connection Error" as a learning opportunity. As you build these milestones at DecodeLabs, you are creating a real-world portfolio that showcases your backend depth to future employers. Your journey to becoming a professional engineer begins right here, right now, with the very first table you define today.



THANK YOU

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